Problem Summary

Given a string containing, determine if it is a palindrome

Rules:

→ ignore non-alphanumeric characters

→ ignore upper and lowercase

A palindrome reads the same forward and backward

Key Idea

Use two pointers:

→ left starts at beginning

→ right starts at end

Move inward:

→ Skip characters that are not letters or numbers

→ Compare lowercase versions of valid characters

If any pair does not match, return False

If all pairs match, return True

Diagram Example

Non valid string: "Race a car"

raceacar

Not same backwards/forward

Valid string: "A man, a plan, a canal: Panama"

amanaplanacanalpanama

Same forward and backwards

Approach

1) Set left pointer at start

2) Set right pointer at end

3) while left < right:

→ Skip invalid characters

→ compare lowercase characters

→ if mismatch, return False

4) If loop finishes returns True

Time Complexity: O(n) → each character is checked at most once

Space Complexity: O(1) → only pointers are used